Torque Control Work Instruction - ECU Bracket Assembly

Document ID	WI-ASSY-001
Area	Assembly Line A3
Revision	R2
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Language	English

1. Purpose

Define the standard tightening and traceability steps for the ECU bracket assembly at Line A3. The instruction is designed to test native PDF text extraction, numeric spec retrieval, and ordered procedure understanding.

2. Required equipment

- Digital torque tool TQ-17 with valid calibration status.
- Fixture F-ECU-03 and barcode scanner SC-09.
- M6 flange bolts, material code BLT-M6-25.

3. Critical specifications

Characteristic	Specification	Reaction if out of spec
M6 bolt torque	9.5 ± 0.5 N·m	Stop station; re-check tool and quarantine unit
Bolt sequence	1 → 3 → 2 → 4	Repeat tightening in correct sequence
Traceability scan	Part + tool result required	Do not release unit without successful scan

4. Procedure

1. Confirm the correct part number and fixture before loading the bracket.
2. Seat four M6 bolts by hand until the bolt heads contact the bracket.
3. Tighten in sequence 1-3-2-4 using TQ-17.
4. Verify each recorded torque is between 9.0 and 10.0 N·m.
5. Scan the unit barcode; confirm the tool result uploads successfully.
6. If any torque is outside range, place the unit in the red quarantine tray and notify the team leader.